Query Processing improvements in SQL Server 2025

Improve the performance of existing workloads!

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Slide and demo: bit.ly/DataSatPordenone2026-sgovoni

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Agenda

- Intelligent Query Processing (IQP)
  - Introduction
  - What’s new in SQL Server 2025 and Azure SQL

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Introduction to Intelligent Query Processing (IQP)

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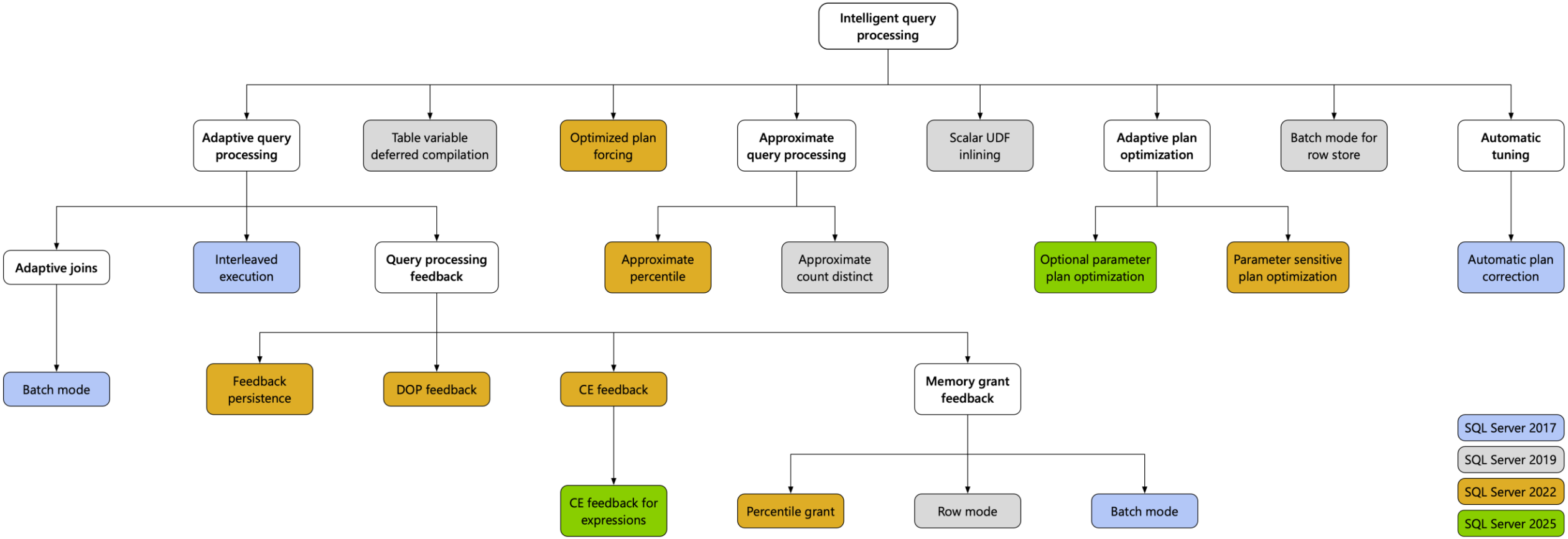
Introduction to IQP

- A set of advanced features introduced to improve performance and reliability of existing workloads
- Designed to work automatically with minimal or no code changes required
- It was first introduced in SQL Server 2017
- It has been enhanced with additional features in subsequent versions, including SQL Server 2019 and later

Introduction to IQP



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What’s new in IQP

For SQL Server 2025 and Azure SQL

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What's new in IQP

- Optional Parameter Plan Optimization (OPPO)
  - The feature can choose a more optimal plan at runtime based on whether a parameter is NULL OR NOT NULL
- Cardinality Estimation (CE) feedback for expressions
  - Extends CE feedback to improve cardinality estimates for repeating expressions across queries
- OPTIMIZED_SP_EXECUTESQL
  - It reduces the impact of compilation storms

IQP features retired during public preview

- Optimized Halloween protection, reduces tempdb space consumption and improves query performance by redesigning the way the database engine solves the Halloween problem

- Due to a potential data integrity issue that might occur if optimized Halloween protection is enabled, Microsoft has **temporarily removed** this feature from SQL Server 2025

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Optional Parameter Plan Optimization (OPPO)

Optional Parameter Plan Optimization

- It improves query plan quality for queries that include optional parameters
- The term “optional parameters” refers to a specific variation of the parameter sensitivity problem
- It uses the adaptive plan optimization (multiplan) infrastructure introduced with PSP optimization

Optional Parameter Plan Optimization

- Consider a predicate like this `((Column = @P1) OR (@P1 IS NULL))`
- When OPPO is **not enabled**, SQL Server always selects an execution plan that performs a table scan ☹
- Using OPTIMIZE FOR may also be ineffective
- Currently, there is no operator that can dynamically convert an index seek into a scan

Optional Parameter Plan Optimization

- Consider a predicate like this `((Column = @P1) OR (@P1 IS NULL))`
- When OPPO **is enabled**, it uses the multi-plan infrastructure from PSP optimization to generate multiple execution plans for a single query ☺
- The optimizer to evaluate different plans based on parameter values and choose the most suitable one at runtime

OPPO limitations and availability

- OPPO applies only to queries eligible for multiplan optimization
- The feature is not applied in scenarios that include
  - Queries using local variables instead of parameters
  - Queries compiled with OPTION (RECOMPILE)
  - Queries executed with SET ANSI_NULLS OFF
  - Statements with automatic parameters

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SQL Server Cardinality Estimation (CE)

The evolution of CE up to CE feedback for expressions

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Introduction to SQL Server CE

- What is the Cardinality Estimator?
  - A core component of the SQL Server query engine that estimates the number of rows (cardinality) each operator will output during query execution

- Why is it important?
  - Cardinality estimates influence the choice of operators
  - Essential for generating efficient query execution plans

Introduction to SQL Server CE

- CE analyzes table and index statistics to predict row counts at each step of query processing
- CE uses statistical models and assumptions
- Before SQL Server 2014: Legacy CE with less accurate estimates on complex queries
- SQL Server 2014 and later: Updated CE with improved models for better accuracy

SQL Server CE feedback

- CE feedback was introduced as part of IQP enhancements in SQL Server 2022
- Enables SQL Server to automatically learn from actual query executions and **adjust** cardinality estimates dynamically
- Reduces performance issues caused by inaccurate cardinality estimates
- CE feedback operates **transparently** with no changes required to application code

SQL Server CE feedback for expression

- CE feedback for expressions extends the framework started by the CE feedback feature
- It learns from previous executions of expressions across queries, to find appropriate CE model choices and apply what has been learned to future executions of those expressions
- Model recommendations are tested and applied automatically to future query executions

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CE feedback limitations and availability

- CE feedback is available only in the Enterprise edition
- CE feedback for expressions is applied differently on primary and secondary replicas; however, on secondary replicas, the feedback is not persisted and exists only in the memory cache

Summary and resources

Summary

- SQL Server 2025 brings key enhancements to the database Engine for better performance and scalability
- Intelligent Query Processing adds new features to optimize workloads with minimal changes, in particular:
  - Optional Parameter Plan Optimization improves query plan quality for queries that include optional parameters
  - CE feedback for expressions extends the framework started by the CE feedback feature

Resources

- Intelligent query processing in SQL databases
- Intelligent query processing features in detail
- Optional parameter plan optimization (OPPO)
- Better plans with Optional Parameter Plan Optimization in SQL Server 2025
- Cardinality estimation (CE) feedback
- The Problem of Cardinality Estimation
- Editions and supported features of SQL Server 2025

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Q&A

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Thanks!